

- 2 (a) (i) x_0 is the decrease in height of the load and is the extension of the spring when the load is at the equilibrium position.

decrease in G.P.E. = mgx_0

Solution 1

magnitude of the external force at the equilibrium position, $F = mg$

$$\text{increase in E.P.E.} = \frac{1}{2}Fx_0 = \frac{1}{2}mgx_0$$

Solution 2

$$\text{increase in E.P.E.} = \frac{1}{2}kx_0^2 = \frac{1}{2}(kx_0)x_0 = \frac{1}{2}mgx_0$$

where k is the spring constant and $kx_0 = mg$ at equilibrium

comparing the expressions for G.P.E. and E.P.E.,
decrease in G.P.E. = $2 \times$ increase in E.P.E. (shown)

- (ii) The decrease in G.P.E. is greater than the increase in E.P.E. as there is negative work done by the external force in lowering the load slowly.

- (b) (i) at equilibrium, $kx_0 = mg \Rightarrow k = \frac{mg}{x_0}$ where k is the spring constant

$$\begin{aligned} \text{E.P.E. at lowest point} &= \frac{1}{2}ke^2 \\ &= \frac{1}{2}\left(\frac{mg}{x_0}\right)(2x_0)^2 \\ &= 2mgx_0 \end{aligned}$$

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- (ii) 1. G.P.E. = mgh where h is the distance from the lowest point
G.P.E. varies linearly with distance.

at lowest point: G.P.E. = 0

at highest point: G.P.E. = $mg(2x_0) = 2mgx_0$

at equilibrium position: G.P.E. = mgx_0

2. E.P.E. = $\frac{mg}{2x_0}e^2$ where e is the extension of the spring and the distance from the highest point.
E.P.E. and distance follows a quadratic relationship.

from (b)(i), at lowest point: E.P.E. = $2mgx_0$

at highest point, spring is at its natural length: E.P.E. = 0

at equilibrium position: E.P.E. = $\frac{1}{2}mgx_0$

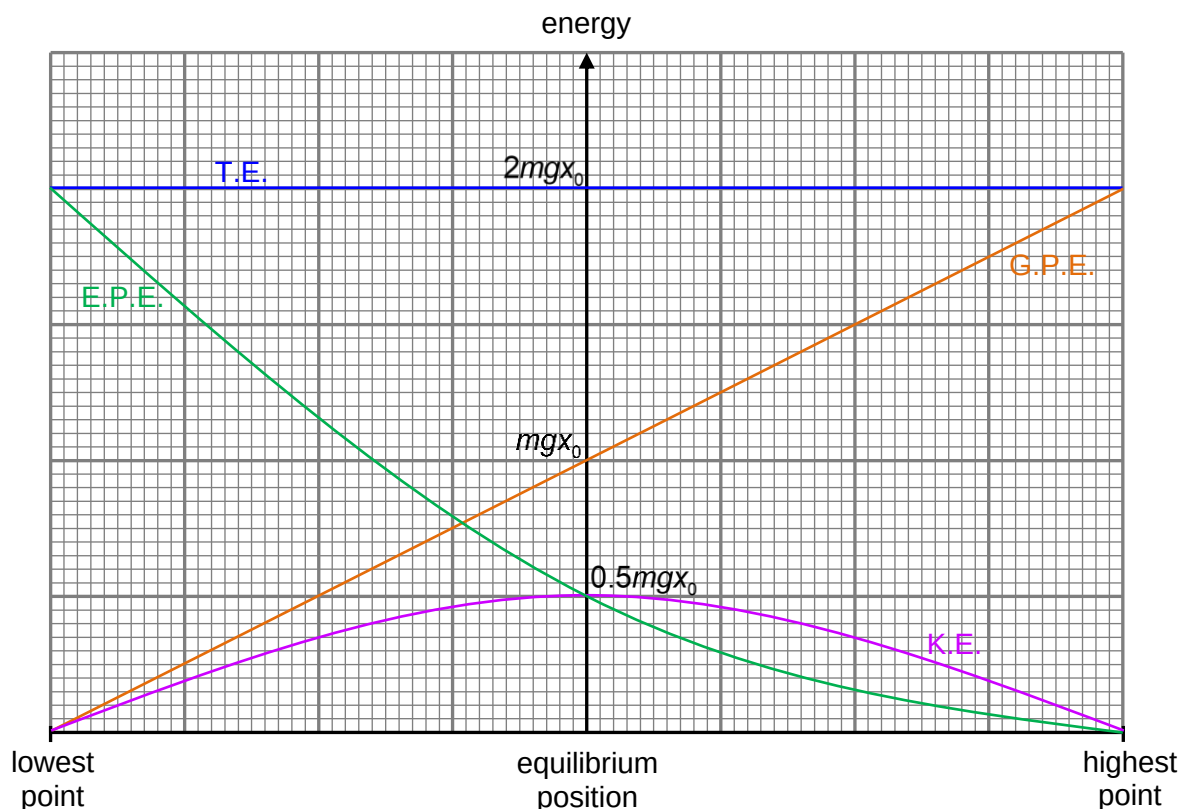
3. K.E. = $\frac{1}{2}m(\pm\omega\sqrt{x_0^2 - x^2})^2 = \frac{1}{2}m\omega^2(x_0^2 - x^2) = -\frac{1}{2}m\omega^2x^2 + \frac{1}{2}m\omega^2x_0^2$
K.E. and displacement follows a quadratic relationship.

at lowest and highest points: K.E. = 0

at equilibrium position:

$$\text{max. K.E.} = \frac{1}{2}m\omega^2x_0^2 = \frac{1}{2}m\left(\frac{k}{m}\right)x_0^2 = \frac{1}{2}\left(\frac{mg}{x_0}\right)x_0^2 = \frac{1}{2}mgx_0$$

4. Total energy at any position is the sum of the G.P.E. , E.P.E. and K.E. at that position.
Total energy remains constant for a system with no energy losses.



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